

Carson Ball

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Education

University of Georgia

B.S. in Computer Systems Engineering | GPA: 3.42

Athens, GA

Expected May 2028

Zell Miller Scholar

Coursework: Design of Digital Systems (Verilog/FPGA), Digital Logic Design, Electronics I, Circuit Analysis, Linear Systems, Systems Programming, Data Structures

Technical Skills

Languages: Verilog, SystemVerilog, C, C++, Python, RISC-V assembly, Bash, MATLAB, Java

Digital Design: RTL design, FSMs, testbenches, Vivado, Basys 3 (Artix-7), Verilator, GTKWave, XDC, timing closure

ASIC: OpenLane/OpenROAD, synthesis, place and route, timing analysis, DRC/LVS, SkyWater 130 nm, Tiny Tapeout

Embedded: STM32 (ARM Cortex-M), bare-metal/RTOS, UART, SPI, I2C, PWM, ADC, interrupts, PID, IMU, encoders

Tools and Lab: Git, Linux, GCC, GDB, Make, Valgrind, ST-Link, oscilloscopes, op-amp/transistor circuits, LaTeX

Projects

RISC-V Processor (FPGA and ASIC)

Sep 2026 – Present

Verilog, C, Python, Verilator, Vivado, OpenLane | Personal project, github.com/cmb42038/riscv-cpu

- Designed a single-cycle RV32I CPU in Verilog from the ISA specification; executes all 37 base integer instructions.
- Built a self-checking Python and Verilator test flow that checks every assembly test against a reference simulator.
- Ran compiled C programs on a Basys 3 (Artix-7) FPGA with UART output, meeting timing through the full Vivado flow.
- Took the design from RTL to GDSII with the OpenLane ASIC flow (SkyWater 130 nm); closed timing, clean DRC/LVS.
- Submitted a UART core to Tiny Tapeout for fabrication on SkyWater 130 nm (silicon expected 2027).
- Extending the design to a 5-stage pipeline with forwarding and hazard detection (in progress).

IEEE Micromouse Robot

Sep 2026 – Present

C/C++, STM32 (ARM Cortex-M) | Sub-team Lead, IEEE UGA Student Branch competition team

- Develop bare-metal C/C++ firmware for an STM32 microcontroller on an autonomous maze-solving robot.
- Own the team's flood-fill algorithm and implement weighted Dijkstra path planning alongside PID motor control.
- Integrate IR distance sensors, wheel encoders, and an IMU for wall detection, odometry, and heading correction.
- Configure STM32 peripherals (PWM timers, ADC, I2C, SPI, UART, interrupts) for motor drive and sensing.

FPGA Digital Design Labs

Aug 2026 – Present

Verilog, Vivado, Basys 3 | CSEE 4270 – Design of Digital Systems

- Designed and verified combinational and sequential RTL, including a Thunderbird turn-signal FSM, through the full Vivado flow (simulation, synthesis, implementation, bitstream).
- Built a behavioral clock divider (100 MHz to 1 Hz) and integrated it with the FSM in a hierarchical top-level module, replacing a bouncing switch-driven clock with the board's oscillator.
- Wrote testbenches to verify waveforms before synthesis, and XDC constraints mapping I/O to switches, LEDs, and clock.

Systems Programming Projects

Jan 2026 – May 2026

C/C++, Linux | CSCI 1730 – Systems Programming

- Built a multithreaded POSIX threads program searching grids up to $3,567 \times 3,567$, comparing 1–3 thread runtimes.
- Wrote a file-comparison tool with low-level Unix system calls, analyzing the speed of byte-by-byte versus bulk reads.
- Compressed DNA sequences to 2 bits per base with bitwise operators, cutting storage about 75% versus ASCII.
- Managed dynamic memory with malloc and free, verifying leak-free programs with Valgrind and debugging with GDB.

Experience

Senior Student Assistant

May 2025 – Present

University of Georgia Libraries | promoted from Student Assistant, Dec 2025

Athens, GA

Host

Jul 2026 – Present

Carrabba's Italian Grill

Athens, GA

Team Member

Jun 2022 – May 2025

Jersey Mike's Subs

Bethlehem, GA

Activities and Honors

IEEE UGA Student Branch (Micromouse sub-team lead) | Eagle Scout